

(A) ENGINEERING DESIGN PROCESS ①

Engineering Design

Engineering design can broadly be classified into two: Invention (novel) design and Improvement design. Invention designs are the types of design that do not exist before while improvement designs modification of an existing design or products. Both types of design are carried out to address some problems or to meet human need. Design activity is not instantaneous process, it occurs over a period of time following a methodological fashion. Engineers are Problem Solvers and what distinguishes a design

Problem from other Problem Solving techniques is the nature of the Problem and the Solution. Design Problems are open-ended and therefore, they have more than one Correct Solution. Unlike analysis Problem that has exact Solution, design Problem by virtue of its Vague nature has more than one Solution.

Design Problems are often solved iteratively. This implies that, as the Solution to the design Problem evolves, the engineer may find himself continually refining the design. A good Solution to a design Problem requires a methodological approach.

Design Process

Design Problems are often solved using five basic steps. By virtue of the Vague nature of design Problems and their multifarious Solution, steps employed for solving design Problems are:

- 1) Problem definition
- 2) Permeant information gathering
- 3) Generation of multiple Solutions
- 4) Analysis and Selection of a Solution
- 5) Testing and implementing the Solution

1) PROBLEM DEFINITION ✓

The solution to a problem involving design begins with a clear and unambiguous definition of the problem. The definition of a design problem evolves through a systematic process as the designer deeply understands the problem. These systematic processes are highlighted below:

- a) Identify and establish the need for the design
Irrespective of its type, design is primarily targeted at solving a problem for the benefit of human beings. Therefore, the engineer ought to recognize a need for a new design or an improvement of a product before he develops a problem definition statement for the design problem.

Although engineers are problem solvers, more often than not, market forces generally establish the need for the design. Sometimes an apparent need may not be the real need. Hence, generating a solution for the apparent need is a wrong approach of solving a design problem. Therefore, solutions should be generated for well-defined real needs.

- b) Develop a Problem Statement

The moment a need for a design has been established, the next thing is to define that need in terms of engineering design problem statement. The problem statement should specifically address the real need but at the same time broad enough to allow wide range of alternative solutions before focusing on specific solution.

- c) Establish Criteria for Success

Criteria for success are the specifications a design solution must meet or the attributes it must possess to be considered successful. The design criteria at this stage of the design process are considered preliminary therefore, they should be flexible throughout the design process. In this way, the initial criteria can easily be refined or modified. It is worthy to note that specifying a sound criteria as a function of the engineer's background knowledge and the industry as

sorter out in that field.

Each Criteria should be evaluated with respect to its applicability to the design effect. Those Criteria that have less significance is the evaluation of the design solution should be excluded. (3)

2) Criteria RELEVANT INFORMATION

Before the generation of alternative solutions, the engineer needs to gather relevant information to the problem at hand. The importance of information gathering of the problem being addressed cannot be overemphasized. Information gathering may reveal some facts about the problem that may necessitate the designer to redefine the problem. Furthermore, mistakes and errors of previous designers can be brought to fore through adequate information gathering.

A good designer needs to ask the following questions although questions may be irrelevant.

- i) Is the problem real and its statement accurate?
- ii) Is there really a need for a new solution or has the problem already been solved?
- iii) What are the existing solutions to the problem?
- iv) What is wrong with the way the problem is currently being solved?
- v) What is right about the way the problem is currently being solved?
- vi) What are the economic factors governing the solution?
- vii) How much will people pay for a solution to the problem?
- viii) What other factors are important to the problem solution? (Such as safety, aesthetic and environmental issues)?

3) GENERATION OF MULTIPLE SOLUTIONS

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In this stage, the designer should start with the existing solutions and find what is wrong with those solutions and focus on the ways to improve the weaknesses of those solutions. This can however be realized by considering new ideas, techniques, and methods to produce a completely unique solution to the problem. This process is referred to as synthesis.

Solutions to engineering design problems do not magically surfaced. Generation of ideas has a lot to do with free thinking and risk taking. Multiple solutions to the problem are often generated by brainstorming.

4) ANALYSIS AND SELECTION OF SOLUTION

The moment alternative solutions to a design problem have been conceived, the solutions need to be analysed in order to appraise the best solution for implementation. Analysis is the result of the design process. It is worthy to note that the of the design process,

a) Analysis of design solutions: - The designer needs to analyse each alternative solution against the selection criteria mentioned above before deciding to select the design solution to implement. Several types of analysis need to be performed on each of the design solution. The uniqueness of every design problem requires different types of analyses. Depending on the nature of the problem and the solution, the following are list of analyses that may need to be considered:

- i) Functional analysis
- ii) Economic
- iii) Manufacturability
- iv) Product safety and liability
- v) Economic and market analysis
- vi) Regulation and Compliance etc.

— functional analysis: — Functional analysis is ⁽⁵⁾ crucial in ascertaining the success of the design solution. The analysis will reveal whether the design solution functions the way it should. A design solution is considered a failure if it does not function properly even if it has met all other criteria.

1 — Ergonomics: — Ergonomics is the study of the interaction between the human body, products and the environment. It is the key factor in the design of all products from furniture machines, and they may provide a source of power or control or even as a sensor in some designs. Therefore, a design solution is considered successful if people are comfortable using it. For example, the handle of a power saw must fit the hand of everybody using it.

— Environmental factors also influence ergonomics. These environmental factors are:

- i) Climate
- ii) Vibration
- iii) Noise
- iv) Illumination

To enhance the comfort and productivity of people operating

mechanical systems, the designs should be mindful of anthropometric data of human beings. Anthropometrics is the use of body measurements to determine the optimum size for products for comfortable and efficient use.

— Examples of anthropometric data are:

- i) How far people can see
- ii) How much space people need
- iii) How much force they can exert
- iv) Height of a person
- v) Length of arms
- vi) Length of legs, etc

A successful design needs to be evaluated and analysed against the distribution of geometry of the people using it.

✓ Product Safety and Liability: — The Primary Consideration for Safety is Product design so to ensure that the Product does not harm human beings. Safety and liability issues may sometimes go beyond human injury to include environmental damage as well as Property damage. Engineers must consider safety in the design in order to avert liability to may arise as a result of unsafe Product. Therefore, designing safety into the Product is the guaranteed way to avert injury. In an event where safety cannot be design into the Product, the second method of designing for safety should be adopted. This method ensures the provision of adequate protection for users of the Product. Protection devices encompasses shield placed around moving and rotating parts, crash protective structures used in vehicles, and 'kill' switches that automatically turn a machine off (as is in considering safety is to ensure that adequate warning labels describing inherent dangers in the Product are placed where necessary.

✶ Product Liability Suit may be the result of a Personal injury due to the operation of a Particular Product. The manufacturer and designer of a Product can be found liable to compensate a user for losses incurred during the operation or use of that Product. To protect themselves is a Product liability trial, Engineers must use State-of-the-art design procedures during the design process.

— Economic and Market Analysis: — The purpose of most Engineering designs is to produce a Product that generates a Profit for the company. Therefore, each alternative features, Potential market, Cost of manufacturing, advertising etc.

Market analysis involves applying Principles of Probability and Statistics to determine if the response of a selected group of People represents the opinion of C.M.

(7)

— Mechanical / Strength Analysis! — The engineer - Conducts mechanical analysis to answer questions such as, 'will the device or structure support the maximum loads that will be subjected to? The effect of shocks and repetitive or dynamic loading over the life of the product must be determined. Many systems generate heat, so the engineer should determine if the system is capable of dissipating the heat generated during normal operation.

The Decision Process

After analysing the alternative solutions, the engineer needs to decide and document which design solution is the best. At this stage, the designer should evaluate each solution objectively against the stated design criteria or requirements. One commonly used method for formalizing the decision-making process is the decision-matrix. The decision-matrix is a mathematical tool that is used to derive a number that specifies and justifies the best decision.

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TEST AND IMPLEMENT THE SOLUTION

This is the final phase of the design process which entails test, construction, and manufacturing of the solution to the design problem. At this stage, designers must consider several methods of implementation, such as prototyping and concurrent engineering. Couple with distinct activities that occur during implementation, such as documentation and applying for patents.

— Prototyping! — The purpose of prototype is to test the design solution under real conditions. Likewise, the designer should build a prototype of the product and subject it to a test under realistic scenario.

— Concurrent engineering! — Concurrent engineering is the parallel implementation of design and analysis in which safety, manufacturability, serviceability, maintainability, and compliance issues are considered early on during the design process. The application of computer-aided design (CAD) make it possible and easier.

to carry out Concurrent engineering during the design process. With the aid of CAD Software, an engineer can design a product and test its functionality and make necessary modifications on the model of the product until he is satisfied that the product meets the desired performance and safety criteria.

The next stage in Concurrent engineering is called 'Art-to-Part'. Here the three-dimensional computer model of the finished design is used with computer-aided manufacturing (CAM) Software to derive appropriate machinery to physically create the Part. Therefore design cycle is therefore almost perfect. Engineer can therefore design the Prototype of a Product in shorter time compare with Serial design. Because of the iterative nature of design, Concurrent engineering significantly shortens the time between iterations. Employing Concurrent engineering, a Product can get to the market much quicker, at a lower cost, and with a higher quality.

— Documentation: — Documenting your work is one of the important activities in design. A designer should be able to communicate the solution to the design problem so that someone else can understand what he has created. This is usually done using technical report.

Communicating the solution of a design problem can be written or both written and oral. Some of the people the designer may be communicating with them are non-technical people and therefore, the ability of a designer to effectively communicate the design solution to non-technical people is paramount. To this effect, the designer is expected to possess a critical skill to enable him excellently communicate and sell a design solution. Graphs, Charts, and for quick and clear presentation.

— Applying for Patent: — A designer that develops a novel or original solution to a design problem should apply for a Patent. A Patent will not protect you from someone else copying your solution, but it does give you specific rights to make and sell your design for a specified period of time.